Performance Estimation Problems (PEPs)

Systematic, principled, and computer-aided approaches to the analysis and design of first-order optimization algorithms

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SMAI-MODE tutorial – March 2026

Organization and learning outcomes (1/3)

Observations:

- ◇ Complexity analyses / convergence proofs for first-order optimization methods often follow very similar (obscure?) patterns.
- ◇ But these proofs are rarely intuitive (rely on combining inequalities).
- ◇ Many variations around same themes.

Answer: *Performance Estimation framework*:

- ◇ help understand proof structures of worst-case analyses,
- ◇ provide principled way to explore complexity bounds/derive such analyses,
- ◇ mathematical proofs via numerical experiments and symbolic computations.

Organization and learning outcomes (2/3)

This tutorial:

- ◇ Interactive introduction to *performance estimation problems*.
- ◇ Learn how to use numerical PEPs to assess performance of simple first-order algorithms.
- ◇ Learn how to exploit PEP outputs to construct Lyapunov-based convergence proofs.
- ◇ Symbolic computations for search & verifying obtained guarantees.
- ◇ Learn the basics of PEPs for numerical design of optimization algorithms.

Organization and learning outcomes (3/3)

0. Organization (now) ($\sim$ 10 mins)
1. Motivations, and a full (simple) example ($\sim$ 90 mins)
2. Missing details (and more!): ($\sim$ 90 mins)
   - first-order oracles, duality, interpolation,
   - semidefinite programming.
3. Guided hands-on notebook examples and Q&A ($\sim$ 60 mins)
4. Compact convergence & non-convergence certificates ($\sim$ 90 mins)
   - Using PEPs to search for cycles (no convergence).
   - Using PEPs to search for Lyapunov functions (convergence).
5. Guided hands-on notebook examples and Q&A ($\sim$ 90 mins)
6. Numerical design of optimized algorithms ($\sim$ 60 mins)
7. Guided hands-on notebook examples and Q&A ($\sim$ 60 mins)

Ressources

Ressources to be found at

`https://github.com/PerformanceEstimation/Tutorial-SMAI-MODE`

Please:

- do interrupt with questions!
- We will adapt the pace; program is probably optimistic.